



Course Description	
COURSE NUMBER and NAME	TECH 100 Information Technology Essentials
UNITS	3
LENGTH OF CLASS	8 Weeks
COURSE DESCRIPTION	Information Technology Essentials is a course focused on providing a comprehensive foundation in major topics of Information Technology and Systems. Topics in computer hardware and software, databases and database management systems, networking, security, emerging technologies, the Internet, and business process management are surveyed.
REQUIRED TEXT	None. Materials and resources will be provided in the course as appropriate.
INSTRUCTIONAL METHOD	Online / On-Campus

Summary of Graded Work and Assessments

Graded work and assessments offer students the opportunity to show the degree of mastery for each CLO. The following table shows how assessments and CLOs align (link).

Assignments	Totals	Weight	CLOs
Engagement and Professionalism (Rubric)	160	16%	1, 2, 3, 4, 5, 6, 7
Week 1 Discussion	30	3%	4, 7
Week 2 Discussion	30	3%	1, 2, 5, 6
Week 2 Assignment	100	10%	3, 4, 5, 7
Week 3 Discussion	30	3%	3, 4, 7
Week 3 Assignment	100	10%	3, 4, 5, 7
Week 4 Discussion	30	3%	1, 2, 3, 4, 7
Week 4 Assignment	100	10%	1, 5
Week 5 Discussion	30	3%	1, 2, 3, 4
Week 5 Assignment	100	10%	1, 5
Week 6 Discussion	30	3%	1, 2, 7



Assignments	Totals	Weight	CLOs
Week 6 Assignment	100	10%	1, 2
Week 7 Discussion	30	3%	1, 2, 7
Week 7 Assignment	100	10%	1, 2, 3, 4, 7
Week 8 Discussion	30	3%	1, 2, 7
Total Points/Percentage	1000 Points	100%	

Course Policies

For Westcliff's course policies, please see the [Course Policies](#) document.

Discussion Requirements

For all discussions, the primary response is due by Wednesday at 11:59 p.m. Pacific Time. The primary response must be at least 200 words in length and fully address the topic, demonstrating critical thinking and understanding. Each student must then also post a minimum of two responses to other students in the discussion by Saturday night at 11:59 p.m. Pacific Time. Each peer response must be at least 50 words in length and substantively engage with the other student's original post, continuing the discussion in a professional manner. If at any time information or material is brought in from an outside source or website, it must be properly cited following APA 7th edition guidelines, and a full reference must be provided.

Assignment Requirements

Each assignment deliverable is specifically defined in the assignment instructions, such as page length, citations and references, audio or video, presentations, tables, etc. For all written assignments, the required page length does not include the cover or references pages. Refer to the specific requirements as stated in each assignment, and reach out to your instructor for additional information as needed. All graded submissions are due by Sunday at 11:59 p.m. Pacific Time.

All written work must adhere to APA 7th edition academic formatting requirements including core components such as the cover page, page numbers, headings, citations, 1" margins, paragraph indentations, left alignment, double spacing throughout, and the final references using hanging indents.

Writing Center

The Westcliff University Writing Center is dedicated to providing quality support to students and faculty. From assignment review, to in-class workshops, to dissertation support, to publication



WESTCLIFF
UNIVERSITY

*College of Technology
& Engineering*

TECH 100 Syllabus

help, the Writing Center is committed to empowering individuals to use the written language to articulate and disseminate knowledge.



Course Learning Outcomes (CLOs)

Learning outcomes are statements that describe significant and essential scholarship that students have achieved and can reliably demonstrate at the end of the course. Learning outcomes identify what the learner will know and be able to do by the end of a course – the essential and enduring knowledge, abilities (skills), and attitudes (values, dispositions) that constitute the integrated learning needed for successful completion of this course. The learning outcomes for this course summarize what students can expect to learn, and how this course is tied directly to the educational outcomes of the degree.

Course Learning Objectives (CLOs)	PLOs
1. Summarize essential and foundational knowledge of computer hardware and software.	1
2. Interpret, identify, and analyze user requirements when considering the selection, creation, evaluation and management of computer-based systems.	1, 2, 5
3. Identify and summarize the business units and disciplines in which computer applications may be applied to attain organizational goals.	1, 2
4. Explain the relationship of the business organization as it relates to information systems from an enterprise perspective.	2
5. Develop the skills in verbal and written communication required to summarize business computing and information processing.	1
6. Recognize and analyze ethical issues and/or dilemmas in information privacy and security settings, then propose reasoned courses of action.	1, 3, 5
7. Research and explain the foundations of implementing technology in business.	1, 2, 5



Detailed Course Outline

The following outline provides important assignment details for this course, unit by unit. Students are responsible for all of the assignments given. Please refer to the Detailed Description of Each Grading Criteria in the syllabus for specific information about each assignment.

Week 1

Assignments to complete this week:

Week 1 Live Class Activity

IT leaders manage a large number of technologies including hardware, software, networks, internet connections, cloud, etc. Break into small groups depending on class size and brainstorm a list of different types of technologies that a business uses and manages. Create a list of these items and determine where the greatest IT costs for a business may come from. People may think that computer hardware is the greatest expense, but is this really the case? Discuss your findings with the live class.

Week 1 Discussion ([Rubric](#))

Review a job information website such as: [CIO](#) or [ITResources](#).

Search for the current top IT careers or technical areas. Select two that are of the most interest to you personally. Discuss what these career areas are, why they are of interest to you, and some of the key skills needed to pursue that career.

Week 2

Assignments to complete this week:

- Review the following cybersecurity resource: [Cybersecurity and Infrastructure Security Agency \(CISA\)](#)

Week 2 Live Class Activity

Break into small groups depending on class size and discuss different security risks between hardware (computers, laptops, tablets, phones, mobile technologies), software (programs, operating systems, cloud applications, etc.), and computer networks (wired, wireless, and cellular). Research a few different security issues in each technology area. Discuss your findings with the live class.

Week 2 Discussion ([Rubric](#))

Watch the following video by Professor Messer on phishing: [Professor Messer: Phishing](#).

Discuss 2 different phishing emails you may have received either personally or at work recently. You may check your junk email folder for examples as well. What are the main premises of the phishing emails? What stands out that would help you identify them as phishing scams? Does your business have any particular required actions that you should take if a phishing or other scam email comes through? What are the ethical considerations?

Week 2 Assignment ([Rubric](#))

Select two of the following cybersecurity issues below. Research what they are, how they work, and how to prevent or respond to them if an attack occurs. Discuss the ethical considerations surrounding cybersecurity attacks and the protection of business data. Write your findings in a 2-3 page APA-formatted paper (not including the cover or references pages).

- Ransomware
- Phishing scams
- Impersonation attacks
- Viruses
- Operating system vulnerabilities
- Mobile device vulnerabilities (cell phones, IoT devices)
- Network security attacks (Firewalls, WiFi, personal hotspots, etc.)
- Social Engineering attacks
- Cloud security (data, internet access, identity verification, etc.)



Week 3

Assignments to complete this week:

- Review the following website: [Project Manager](#)

Week 3 Live Class Activity

Every business has internal processes that they follow to do work. For example, a business sells products or services and they have processes that must be followed in order to order resources used to deliver the products or services, pay employees that perform the work, receive and apply payments made by customers, etc. A key term used in the industry is Business Process Management. Break into small groups depending on class size and select a specific business process used that you or your group are familiar with. Identify the various steps involved and discuss how those steps are managed. Identify where there may be room for potential improvement, and discuss how technologies may be used to help manage the business processes. Return and discuss your findings with the live class.

Week 3 Discussion ([Rubric](#))

Following the Project Management Institute (PMI) and the Project Management Book of Knowledge (PMBOK), there are five process groups and 10 knowledge areas. Select one of the 10 knowledge areas and provide a brief description of what the knowledge area is about. What have you done or are you currently doing in your workplace that relates to this knowledge area?

Week 3 Assignment ([Rubric](#))

Write a 2-3 page APA-formatted paper addressing a technology project that has been recently performed or may currently be ongoing in your workplace. If the project is being managed following the principles reviewed within the PMBOK, please describe how this is working. If it is not (or perhaps not formally), address what could be done to better manage the project following the key principles covered in the PMBOK.



Week 4

Assignments to complete this week:

- Review the following website resources:
 - [Azure Cloud Certification Options](#)
 - [AWS Certification Options](#)
 - [Google Cloud Certification Options](#)
- Optional: Review the following video on [Azure Fundamentals Certification](#)

Week 4 Live Class Activity

We live in a connected world that relies very heavily on the Internet. There are a large number of benefits that the use of the Internet provides, as well as a large number of potential concerns or drawbacks. Break into small groups depending on class size and brainstorm different pros and cons of using the Internet. Do you think we could still function today without it? What would be the problems or challenges if we could not? Return and discuss your thoughts with the live class.

Week 4 Discussion ([Rubric](#))

Watch the video [Cloud Computing in 60 Seconds](#).

Based on information learned from this video, select one of the three main cloud platforms (SaaS, PaaS, IaaS) and identify at least two different uses or services available using the platform you have selected. For example, if you select SaaS, you may choose to address services such as Microsoft Office 365 or Salesforce. If you choose IaaS, perhaps you will address server or network functions available through AWS or Azure. If you choose PaaS, perhaps you will address tools available to build applications, APIs, or web interfaces through Quickbase or Azure. Address how and why these tools may be used in a business setting.

Week 4 Assignment ([Rubric](#))

There are several different environments and certifications available in cloud computing. Select either Amazon Web Services (AWS), Microsoft Azure, or Google Cloud, and use the free system [Loom.com](#) to record a brief 2-3-minute presentation discussing what certifications are available for your selected platform. Discuss the requirements and costs to pursue certifications. Lastly, research potential career options that require the certifications you have identified. Your presentation must include both audio and video in conjunction with your content. Submit your Loom recording link to the assignment dropbox.



Week 5

Assignments to complete this week: Programming languages

- Review the following:
 - [“Top Programming Languages”](#)

Week 5 Live Class Activity

Businesses utilize software for many different purposes. Some applications are bought ‘off-the-shelf’ and used for specific purposes. Some are cloud-based and provided as a service. But not all applications completely meet the needs of a particular business function. In those cases, companies may have to decide whether to make a particular application work for them, or write their own software that is specifically designed to meet their needs. This represents a ‘build vs. buy’ scenario. Break into small groups depending on class size, then research and create a brief list of reasons why a business may want to write their own application and what the challenges may be in doing so. Discuss your findings with the live class.

Week 5 Discussion ([Rubric](#))

Programming essentially provides a computer a very detailed set of instructions to accomplish a task. Many programming languages are in use today, and there are also cloud-based systems that will help users create programs. What is the difference between writing a program from scratch using a programming language like C++, Java, or Python, and using a platform as a service such as Squarespace, GoDaddy, Wix, or Weebly? Why would a business or individual choose to use one over the other?

Week 5 Assignment ([Rubric](#))

Select one of the top programming languages from the list presented in the reading for this week. Find a basic tutorial video on YouTube (or another reputable source) on how to program using that language. Following the tutorial, write a very basic code example between 5-10 lines (such as writing a basic welcome sentence to the screen). Submit your written code as a part of your written paper for this week. In your 2-3-page APA-formatted paper, introduce the programming language you selected, where it is most used, and what your written code will do.



Week 6

Assignments to complete this week: Web

- Review the following website creation links:
 - [Squarespace](#)
 - [Wix](#)
 - [Weebly](#)
 - [GoDaddy](#)

Week 6 Live Class Activity

We all use the Internet every day and browse various websites. Break into small groups depending on class size and consider your top 5 websites that you visit on a regular basis. Discuss with the group how each page is designed and functions. What are the features you like? What do you dislike? Create a common list of features/functions within your group and be prepared to discuss with the live class.

Week 6 Discussion ([Rubric](#))

We all browse the internet frequently and explore many different websites and pages. While some websites may appear very simple to follow and are not visually distracting, others may appear much more complex or harder to navigate. Select two different websites that you visit regularly and describe some of the things you like and dislike about the site designs and why. Be specific regarding the look, coloring, font, navigation, etc.

Week 6 Assignment ([Rubric](#))

Use a free website creator such as Squarespace, Wix, Weebly, or GoDaddy, and create a basic web page. Your page must include the following:

- Your name and university email address
- A picture. If you are not comfortable uploading your own picture, find an appropriate image you can use instead
- The degree program in which you are enrolled
- Your top two areas of interest in technology
- Use appropriate colors and font to make your page look professional

Submit the URL for your website to the assignment text box. You do not need to submit it as an attached paper. Your instructor will visit your site to review and grade.



Week 7

Assignments to complete this week: Data Science

- Reading:
 - [Data Science Isn't Your Only Career Option](#)
 - [Python Data Science Handbook](#)

Week 7 Live Class Activity

Massive amounts of information are collected and stored electronically every minute. All of this information is stored in databases, and we use database management systems (DBMS) to control the various databases and how they all interconnect. Break into small groups depending on class size and research how databases are built and work. Create a 2-3 slide presentation outlining basic database structures and uses, and research and identify a DBMS that can be used to manage a business database. Present your findings in the live class.

Week 7 Discussion ([Rubric](#))

Review the required reading for this week. Discuss the future of data science. Where do you see it going in the next 3-5 years? What challenges do you see business and IT facing?

Week 7 Assignment ([Rubric](#))

Review the following video comparing R and Python for data science:

[R vs Python | Which is Better for Data Analysis?](#)

Write a brief summary of the comparison between R and Python, and take a position on which would be the best system for you to use personally and professionally. Address the pros and cons of each system and how they may apply in a business setting. Support your reasoning in a 2-3-page APA -formatted paper and include appropriate citations and references.



Week 8

Assignments to complete this week: AR/VR

- Reading:
 - [How AR and VR are Transforming the Future of Business](#)
 - [Augmented and Virtual Reality](#)

Week 8 Live Class Activity

There are several emerging technologies being developed and making their way in the world today. Break into small groups depending on class size and identify the top five that your team feels are making the biggest impact today. For example, you may look at AR/VR, AI/ML, Cryptocurrency, blockchain, Industry 4.0, etc. Be prepared to discuss your thoughts with the live class.

Week 8 Discussion ([Rubric](#))

After reviewing the two articles in the assigned reading for this week, select an industry or organization that you are interested in and describe how you think AR/VR could best be implemented. What are some challenges in adding AR/VR to that industry/organization? How would including AR/VR help build the business?